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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 10/30/2025 |
| **PI Name:** | Suman, Shubhankar Ph.D. | | |
| **Project Title:** | Senescent Cell Targeting to Alleviate Space Radiation-Induced Neuroinflammation | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders<br>(2) **CNS**:Risk of Acute (In-flight) and Late Central Nervous System Effects from Radiation Exposure | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 20007 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity |
| **Start Date:** | 01/01/2024 | **End Date:** | 12/31/2027 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2027 per J. McFather/NASA JSC (Ed., 5/6/26).<br>NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 12/30/24). | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Li, James Ph.D. ( Georgetown University ) | | |
| **Grant/Contract No.:** | 80NSSC24K0287 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The cerebral cortex, the outermost layer of the mammalian brain, regulates important functions such as awareness, motor and sensory functions, memory, language, and visual perceptions. Earlier studies using animal models have reported that exposure to heavy-ion space radiation triggered persistent DNA damage and neuroinflammation in the cerebral cortex. However, we have yet to clarify how persistent DNA damage and neuroinflammation are sustained long-term after space radiation exposure. Previous research has linked cellular senescence, the acquisition of the senescence-associated secretory phenotype (SASP), and increased senescence induced inflammatory response (SIR) to the onset of neuroinflammation during aging; however, the role of senescent/SASP cells in chronic neuroinflammation after heavy-ion space radiation warrants further investigation. Our central hypothesis is that persistent DNA damage related to senescence and the senescence inflammatory response (SIR) is the primary driver of sustained neuroinflammatory stress in the cerebral cortex following exposure to space radiation. This investigation aims to identify, quantify, and characterize senescence-susceptible cells in the cerebral cortex and evaluate the efficacy of senescent cell ablation (senolysis) as a strategy to mitigate energetic heavy ions (HZE) or simulated galactic cosmic radiation (GCRsim)-induced neuroinflammation. We propose that targeted removal of these senescent cells may reduce the risk of neuroinflammation. This proposal will expand NASA's research portfolio to encompass the implications of senescence and SIR in relation to space radiation-induced CNS effects. Since many senolytic drugs struggle to penetrate the blood-brain barrier, our approach of genetic senescent cell ablation will provide robust proof-of-principle for the feasibility of senolytic strategies to mitigate HZE-induced CNS effects. This study will enhance our understanding of cell-type-specific biological responses in the cerebral cortex and their role in sustained neuroinflammation. By establishing senescence/SIR signaling as a critical contributor to space radiation-induced neuroinflammation, our investigation will lay the groundwork for future interventions, ultimately aiming to alleviate space radiation-induced CNS effects. Aim 1: Identify, quantify, and characterize senescence/SIR-susceptible cerebral cortex cell types. We will analyze the dose- and time-dependent effects of HZE and gamma radiation on oxidative stress, senescence, SIR, and neuroinflammation. Our working hypothesis posits that the high-LET characteristics of 28Si ions will induce a greater activation of senescence/SIR signaling compared to gamma rays. Aim 2: Obtain "proof-of-principle" that senescent cell removal could alleviate HZE-induced persistent neuroinflammation in the cerebral cortex. Our hypothesis is that the selective removal of senescent cells will lead to a reduction in HZE-induced neuroinflammation. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research project aims to advance our understanding of the long-term effects of energetic heavy ions (HZE) and simulated GCR exposures on the CNS, particularly in relation to neuroinflammation and cellular senescence. By identifying the cellular mechanisms that sustain neuroinflammatory stress in the cerebral cortex, we will provide critical insights into the fundamental processes of aging and age-related neurological diseases. Implications of our findings extend beyond space exploration. As populations age on Earth, neuroinflammatory conditions, including Alzheimer's disease and other forms of dementia, pose significant public health challenges. The identification of senescence and the senescence inflammatory response (SIR) as pivotal contributors to sustained neuroinflammation could lead to novel therapeutic strategies that target senescent cells, potentially reducing the burden of these age-associated diseases. Moreover, the exploration of senolytic therapies may pave the way for innovative treatments that enhance brain health and longevity. By providing a proof of principle for senescent cell ablation strategies, our research could catalyze the development of new drugs or interventions aimed at mitigating neuroinflammation and improving cognitive function in aging populations. In summary, this research not only seeks to protect astronauts from the detrimental effects of space radiation but also holds the potential to significantly impact public health by offering new avenues for the prevention and treatment of neurodegenerative diseases on Earth. |
| **Task Progress:** | During the previous year, we studied how exposure to space radiation can cause cortical brain cells to age prematurely and trigger inflammation, both of which are linked to memory and cognitive decline. We tested both genetic and drug-based strategies to remove these "aged" or senescent cells. Mice genetically engineered to eliminate damaged brain cells after radiation exposure showed reduced signs of inflammation and brain cell stress. In concurrence, mice treated with fisetin (a pharmacological senolytic agent) reduce brain inflammation and markers of aging. Imaging studies confirmed changes in brain metabolites caused by simulated space radiation, which were improved after these treatments. These findings suggest that clearing aged brain cells may help protect astronauts from the harmful effects of deep-space radiation on brain health. In the current reporting period (01/01/2025 to 12/31/2025), we focused on identifying mouse cortical cell types most sensitive to space-like radiation. Female mice were exposed to simulated galactic cosmic radiation (GCRsim; 33-ion full spectrum), five months later, brain tissue was examined to see which cells showed signs of aging or inflammation. We focused on three main cell types: neurons (which process information), astrocytes (which support neurons), and microglia (the brain's immune cells). Our results showed that microglia, the brain's immune cells, were the most affected by radiation, showing strong signs of cellular aging. Astrocytes and neurons were also impacted, but to a lesser extent. This means radiation may primarily trigger the brain's immune system, leading to long-term inflammation. We found changes in several key proteins linked to inflammation, oxidative stress, and brain degeneration, similar to what occurs in diseases like Alzheimer's. These changes indicate that space radiation can cause persistent stress in the brain's environment, promoting conditions that may accelerate cognitive decline. This research helps NASA understand how space radiation might affect astronaut's brain health during long-duration missions to the Moon or Mars. We are now testing senolytic drugs like fisetin and ABT-263 that target and remove these aged, inflammation-causing cells. The goal is to see if these treatments can prevent memory and behavioral problems after radiation exposure. Ultimately, this work may lead to new ways to protect astronauts' brains during deep-space travel and could also benefit people on Earth by informing treatments for age-related brain diseases. |
| **Bibliography Type:** | Description: (Last Updated: 11/12/2025) |
| **Abstracts for Journals and Proceedings** | Kumar K, Angdisen J, Ma J, Fornace AJ Jr, Suman S. "Alleviating space radiation-induced neuroinflammation in mice using a genetic and drug-based senolytic approach." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |

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| **Abstracts for Journals and Proceedings** | Kumar K, Angdisen J, Fornace AJ Jr, Suman S. "Profiling of cellular senescence and SASP in mouse cerebral cortex after exposure to simulated galactic cosmic radiation." 71st Annual meeting of the Radiation Research Society (RRS), San Juan, Puerto Rico, September 21-24, 2025.<br>Abstracts. 71st Annual meeting of the Radiation Research Society (RRS), San Juan, Puerto Rico, September 21-24, 2025. , Sep-2025 |
| **Abstracts for Journals and Proceedings** | Suman S. "Potential countermeasure strategies against adverse health effects induced by space radiation (Invited talk)." International Radiobiology Conference on 'Biological Effects of Space Radiation, Heavy Ions & Human Space Missions-Mechanisms & Biomedical Countermeasures,' New Delhi, Delhi, India, February 27-March 1, 2025.<br>International Radiobiology Conference on 'Biological Effects of Space Radiation, Heavy Ions & Human Space Missions-Mechanisms & Biomedical Countermeasures,' New Delhi, Delhi, India, February 27-March 1, 2025. , Mar-2025 |
| **Articles in Peer-reviewed Journals** | Kumar K, Kumar S, Angdisen J, Datta K, Fornace AJ Jr, Suman S. "Radiation quality-dependent progressive increase in oxidative DNA damage and intestinal tumorigenesis in Apc1638N/+ mice. " Curr Oncol. 2025 Jul 1;32(7):382. https://doi.org/10.3390/curroncol32070382 ; PubMed PMID: 40710193; PubMed Central PMCID: PMC12293654 , Jul-2025 |
| **Articles in Peer-reviewed Journals** | Suman S. "Integrative analysis of radiation-induced senescence-associated secretory phenotype factors in kidney cancer progression." Genes (Basel). 2025 Jan 15;16(1):85. https://doi.org/10.3390/genes16010085 ; PubMed PMID: 39858632; PubMed Central PMCID: PMC11765417 , Jan-2025 |